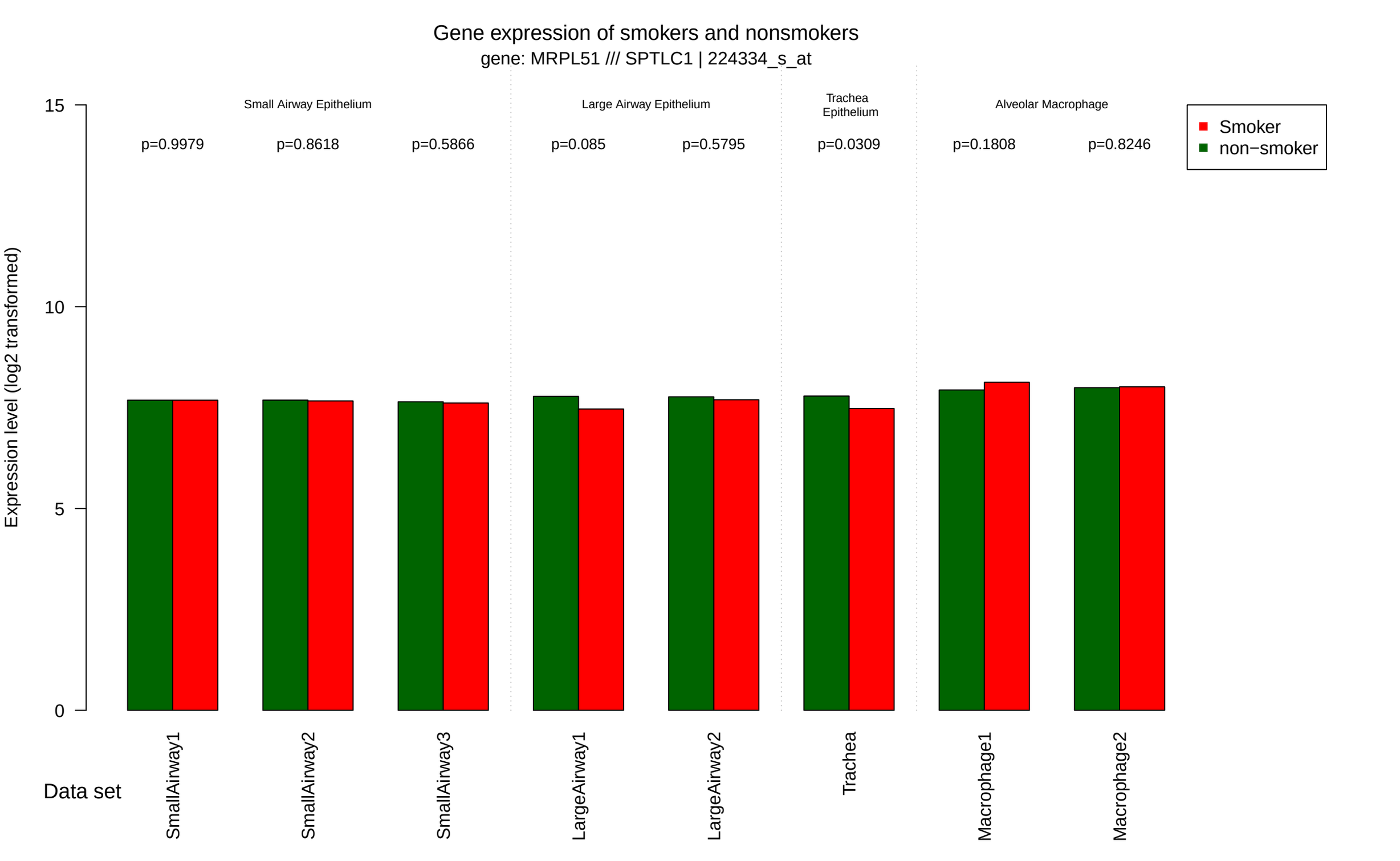
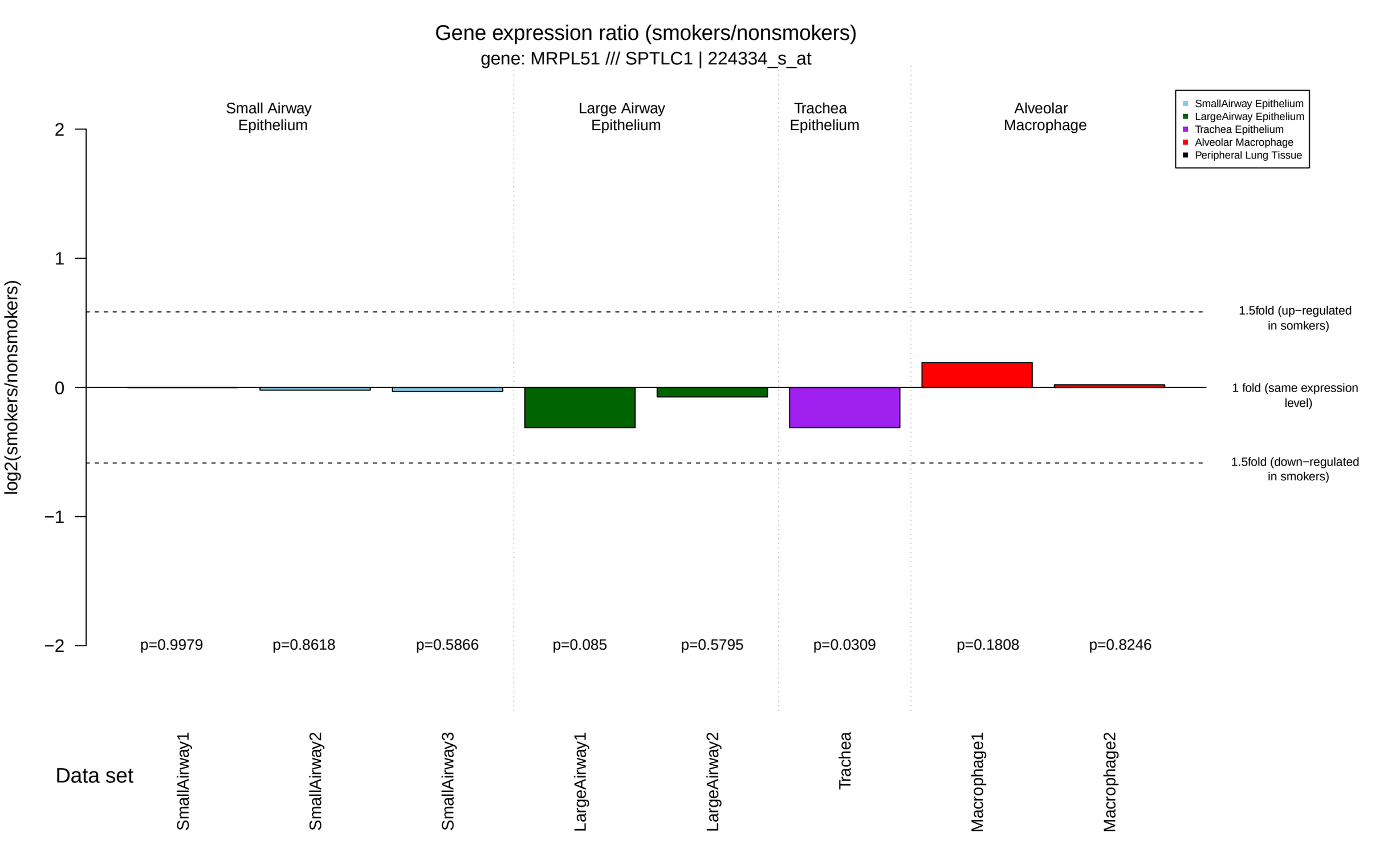
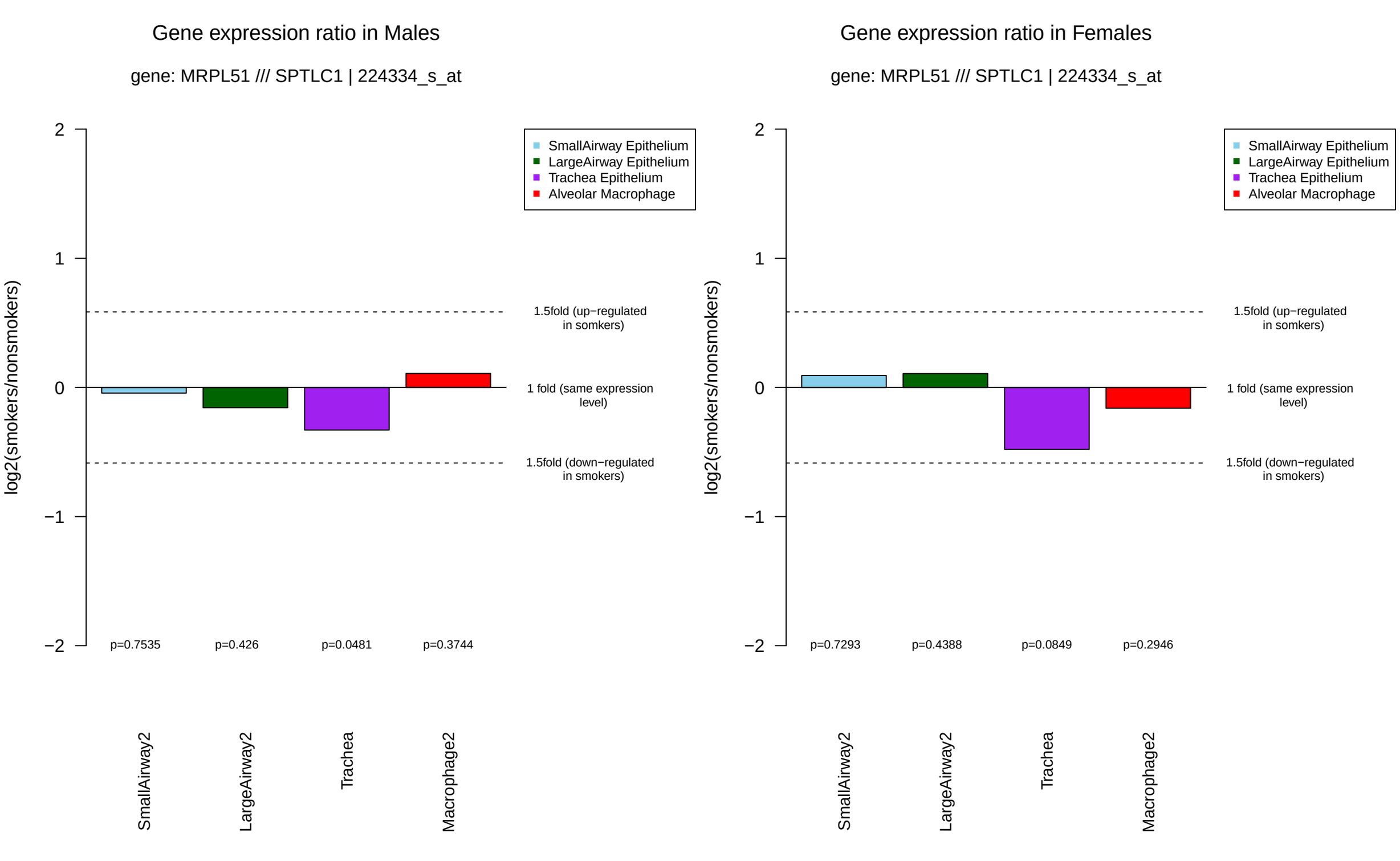
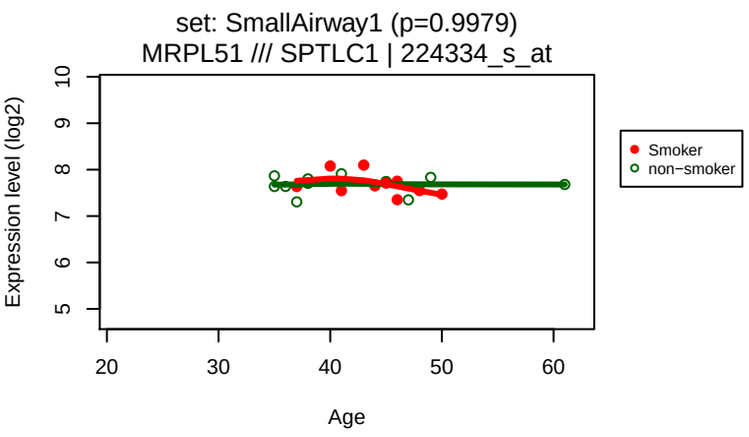
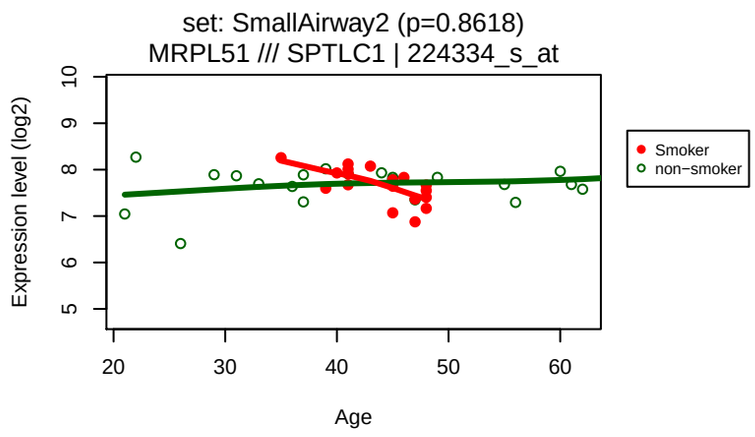
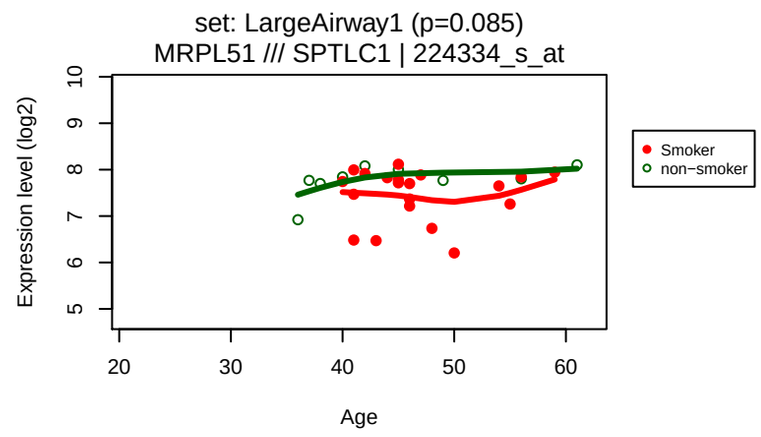

Gene expression of smokers and nonsmokers
gene: MRPL51 /// SPTLC1 | 224334_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9979
p=0.8618
p=0.5866
p=0.085
p=0.5795
p=0.0309
p=0.1808
p=0.8246
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

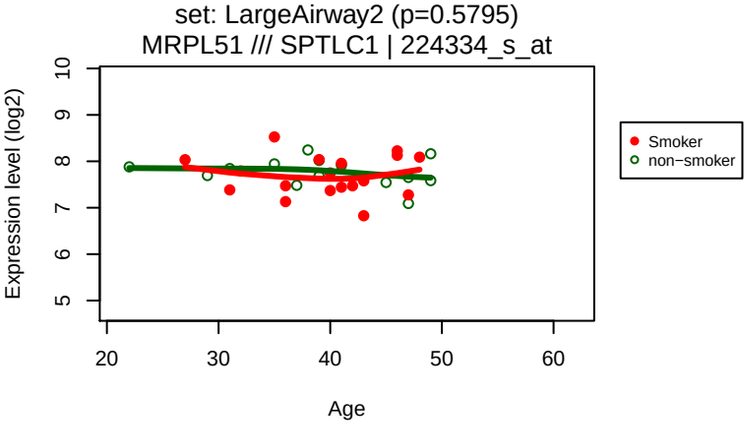

Gene expression ratio (smokers/nonsmokers)
gene: MRPL51 /// SPTLC1 | 224334_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9979
p=0.8618
p=0.5866
p=0.085
p=0.5795
p=0.0309
p=0.1808
p=0.8246
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

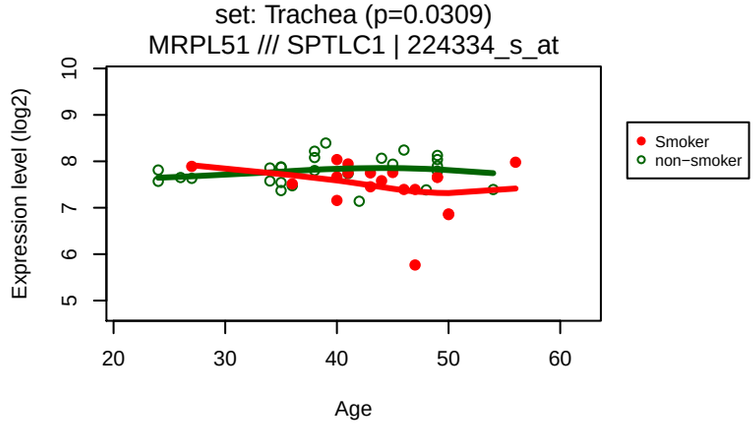

Gene expression ratio in Males
gene: MRPL51 /// SPTLC1 | 224334_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7535
p=0.426
p=0.0481
p=0.3744
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRPL51 /// SPTLC1 | 224334_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7293
p=0.4388
p=0.0849
p=0.2946
SmallAirway2
LargeAirway2
Trachea
Macrophage2

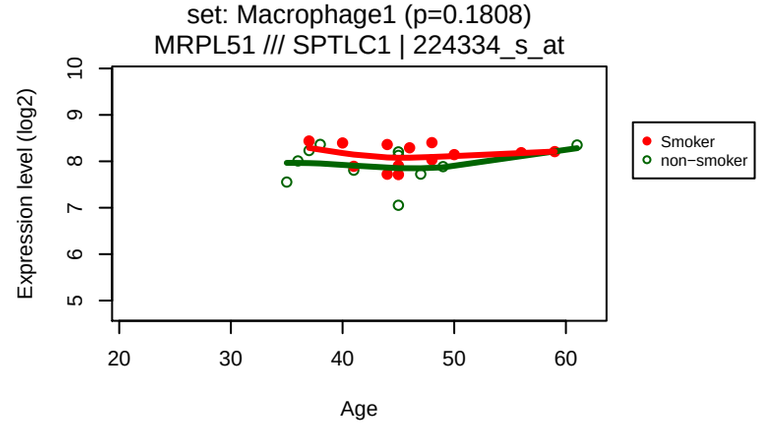

set: SmallAirway1 (p=0.9979)
MRPL51 /// SPTLC1 | 224334_s_at
Expression level (log2)
Age
Smoker
non−smoker

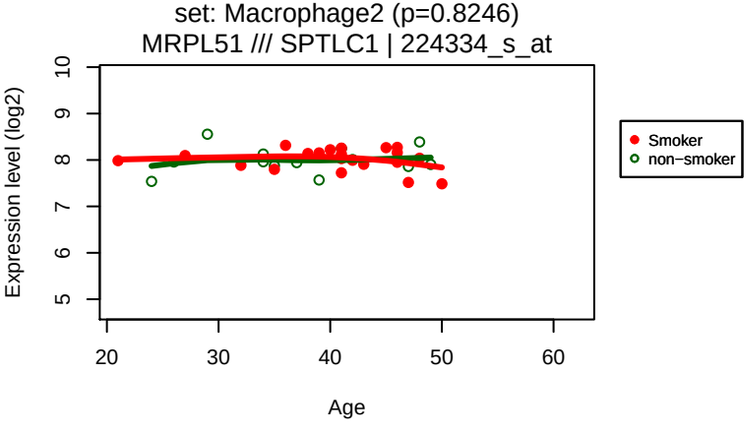

set: SmallAirway2 (p=0.8618)
MRPL51 /// SPTLC1 | 224334_s_at
Expression level (log2)
Age
Smoker
non−smoker

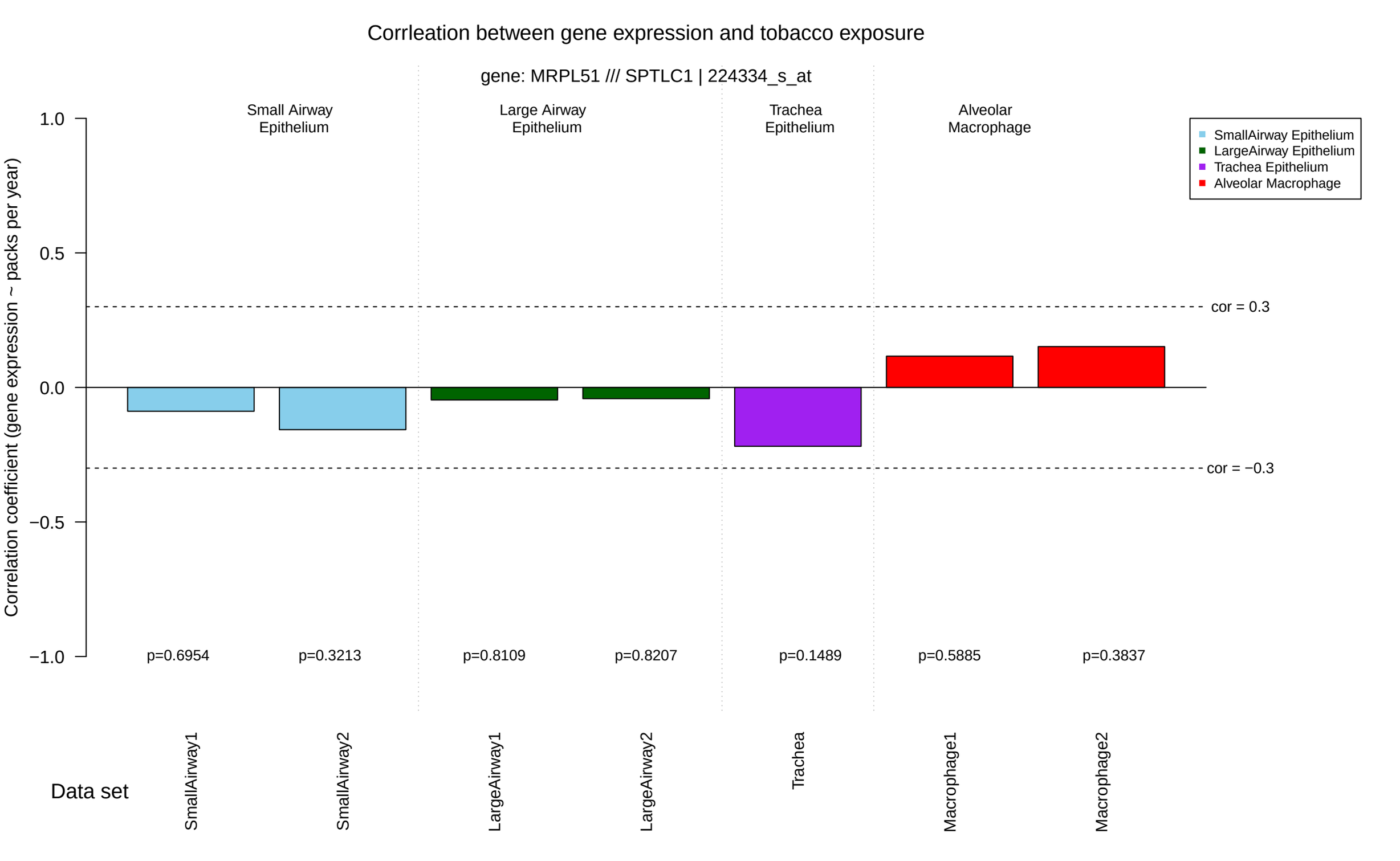

set: LargeAirway1 (p=0.085)
MRPL51 /// SPTLC1 | 224334_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5795)
MRPL51 /// SPTLC1 | 224334_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0309)
MRPL51 /// SPTLC1 | 224334_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1808)
MRPL51 /// SPTLC1 | 224334_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8246)
MRPL51 /// SPTLC1 | 224334_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRPL51 /// SPTLC1 | 224334_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6954
p=0.3213
p=0.8109
p=0.8207
p=0.1489
p=0.5885
p=0.3837
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2